

```
In [48]: !git clone https://github.com/nngasu4course/Face_Recognition.git
```

fatal: destination path 'Face_Recognition' already exists and is not an empty directory.

```
In [49]: !pip install face_recognition
```

Requirement already satisfied: face_recognition in /usr/local/lib/python3.12/dist-packages (1.3.0)
Requirement already satisfied: face-recognition-models>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from face_recognition) (0.3.0)
Requirement already satisfied: Click>=6.0 in /usr/local/lib/python3.12/dist-packages (from face_recognition) (8.3.1)
Requirement already satisfied: dlib>=19.7 in /usr/local/lib/python3.12/dist-packages (from face_recognition) (19.24.6)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from face_recognition) (2.0.2)
Requirement already satisfied: Pillow in /usr/local/lib/python3.12/dist-packages (from face_recognition) (11.3.0)

```
In [50]: # Скачиваем изображения
import urllib
url, filename = ("https://github.com/nngasu4course/Face_Recognition/raw/main/elon.jpg"
try: urllib.urlopen().retrieve(url, filename)
except: urllib.request.urlretrieve(url, filename)
```

```
In [51]: import urllib
url, filename = ("https://github.com/nngasu4course/Face_Recognition/raw/main/Donald_1"
try: urllib.urlopen().retrieve(url, filename)
except: urllib.request.urlretrieve(url, filename)
```

```
In [52]: import urllib
url, filename = ("https://github.com/nngasu4course/Face_Recognition/raw/main/jeffbezo"
try: urllib.urlopen().retrieve(url, filename)
except: urllib.request.urlretrieve(url, filename)
```

Training Images

```
In [53]: import face_recognition
import numpy as np
from PIL import Image, ImageDraw
from IPython.display import display
from google.colab.patches import cv2_imshow
import cv2
from datetime import datetime

face_1 = face_recognition.load_image_file("elon.jpg")
face_1_encoding = face_recognition.face_encodings(face_1)[0]

face_2 = face_recognition.load_image_file("Donald_Trump.jpg")
face_2_encoding = face_recognition.face_encodings(face_2)[0]

face_3 = face_recognition.load_image_file("jeffbezoz.jpg")
face_3_encoding = face_recognition.face_encodings(face_3)[0]

known_face_encodings = [
    face_1_encoding,
    face_2_encoding,
    face_3_encoding
]
known_face_names = [
    "Elon Musk",
    "Donald Trump",
    "Jeff Bezos"
]
print("Done learning and creating profiles")
```

Done learning and creating profiles

add names to the "attendance_list.csv" file

```
In [54]: def makeAttendanceEntry(name):
    with open('attendance_list.csv', 'r+') as FILE:
        alllines = FILE.readlines()
        attendanceList = []
        for line in alllines:
            entry = line.split(',')
            attendanceList.append(entry[0])
        if name not in attendanceList:
            now = datetime.now()
            dtString = now.strftime('%d/%b/%Y, %H:%M:%S')
            #f.writelines(f'\n{name},{dtString}')
```

Face Recognition

```
In [55]: import urllib
url, filename = ("https://github.com/nngasu4course/Face_Recognition/raw/main/unknown_
try: urllib.URLopener().retrieve(url, filename)
except: urllib.request.urlretrieve(url, filename)
```

```
In [56]: file_name = "unknown_do.jpg"
unknown_image = face_recognition.load_image_file(file_name)
unknown_image_to_draw = cv2.imread(file_name)

face_locations = face_recognition.face_locations(unknown_image)
face_encodings = face_recognition.face_encodings(unknown_image, face_locations)

pil_image = Image.fromarray(unknown_image)
draw = ImageDraw.Draw(pil_image)

for (top, right, bottom, left), face_encoding in zip(face_locations, face_encodings):
    # See if the face is a match for the known face(s)
    matches = face_recognition.compare_faces(known_face_encodings, face_encoding)

    name = "Unknown"

    face_distances = face_recognition.face_distance(known_face_encodings, face_encoding)
    best_match_index = np.argmin(face_distances)
    if matches[best_match_index]:
        name = known_face_names[best_match_index]

    # Draw a box around the face using the Pillow module
    cv2.rectangle(unknown_image_to_draw, (left, top), (right, bottom), (0, 255, 0), 3)
    draw.rectangle(((left, top), (right, bottom)), outline=(0, 255, 255))
    cv2.putText(unknown_image_to_draw, name, (left, top - 20), cv2.FONT_HERSHEY_SIMPLEX, 1,
    print(name)
    makeAttendanceEntry(name)

# display(pil_image)
cv2.imshow(unknown_image_to_draw)
```

Donald Trump

